Dynamic Analysis

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{f}$$

$$\mathbf{u}(t=0) = \mathbf{u_0}$$

$$\dot{\mathbf{u}}(t=0) = \mathbf{v_0}$$

Newmark time-integration scheme
$(e.g. \gamma = 0.5 \; and \; \beta = 0.25)$

$$\mathbf{M}\ddot{\mathbf{u}}_{n+1} + \mathbf{K}\mathbf{u}_{n+1} = \mathbf{f}_{n+1}$$

$$\mathbf{u}_{n+1} = \mathbf{u}_n + \Delta t\dot{\mathbf{u}}_n + \frac{\Delta t^2}{2}\left(1-2\beta\right)\ddot{\mathbf{u}}_n + \Delta t^2\beta\ddot{\mathbf{u}}_{n+1}$$

$$\dot{\mathbf{u}}_{n+1} = \dot{\mathbf{u}}_n + \Delta t\left(1-\gamma\right)\ddot{\mathbf{u}}_n + \Delta t\gamma\ddot{\mathbf{u}}_{n+1}$$

$$\left[\frac{1}{\Delta t^2\beta}\mathbf{M} + \mathbf{K}\right]\mathbf{u}_{n+1} = \mathbf{f}_{n+1} + \frac{1}{\Delta t^2\beta}\mathbf{M}\tilde{\mathbf{u}}_{n+1}$$

$$\boxed{\mathbf{u}_{n+1} = \left[\frac{1}{\Delta t^2\beta}\mathbf{M} + \mathbf{K}\right]^{-1}\left(\mathbf{f}_{n+1} + \frac{1}{\Delta t^2\beta}\mathbf{M}\tilde{\mathbf{u}}_{n+1}\right)}$$

where

$$\ddot{\mathbf{u}}_{n+1} = \frac{\mathbf{u}_{n+1} - \tilde{\mathbf{u}}_{n+1}}{\Delta t^2\beta}$$

$$\dot{\mathbf{u}}_{n+1} = \tilde{\dot{\mathbf{u}}}_{n+1} + \frac{\gamma}{\Delta t\beta}\left(\mathbf{u}_{n+1} - \tilde{\mathbf{u}}_{n+1}\right)$$

$$\tilde{\mathbf{u}}_{n+1} = \mathbf{u}_n + \Delta t\dot{\mathbf{u}}_n + \frac{\Delta t^2}{2}\left(1-2\beta\right)\ddot{\mathbf{u}}_n$$

$$\tilde{\dot{\mathbf{u}}}_{n+1} = \dot{\mathbf{u}}_n + \Delta t\left(1-\gamma\right)\ddot{\mathbf{u}}_n$$

Consistent Mass Matrix

1D

$$\mathbf{M}^e = \frac{1}{6}\rho^e A^e l^e \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

2D

$$\mathbf{M}^e = \frac{1}{6}\rho^e A^e l^e \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

3D

$$\mathbf{M}^e = \frac{1}{6}\rho^e A^e l^e \begin{bmatrix} 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 1 \\ 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 2 \end{bmatrix}$$

Computation of Mass matrix

For each bar $e = 1 \ldots n_{el}$

a)Compute element mass matrix

$$x_1^e = \mathbf{x}(\mathbf{T}_n(e,1),1), \qquad x_2^e = \mathbf{x}(\mathbf{T}_n(e,2),1)$$
$$y_1^e = \mathbf{x}(\mathbf{T}_n(e,1),2), \qquad y_2^e = \mathbf{x}(\mathbf{T}_n(e,2),2)$$
$$z_1^e = \mathbf{x}(\mathbf{T}_n(e,1),3), \qquad z_2^e = \mathbf{x}(\mathbf{T}_n(e,2),3)$$
$$l^e = \sqrt{(x_2^e - x_1^e)^2 + (y_2^e - y_1^e)^2 + (z_2^e - z_1^e)^2}$$
$$\mathbf{M}^e = \frac{1}{6} A^e l^e \rho^e \begin{bmatrix} 2 & \cdots & \\ \vdots & \ddots & \vdots \\ & \cdots & 2 \end{bmatrix}$$

b)Store element matrix

For each $r = 1 \ldots n_{ne} \times n_i$

For each $s = 1 \ldots n_{ne} \times n_i$

$$\mathbf{M}_{el}(r,s,e) = \mathbf{M}^e(r,s)$$

Next s

Next r

Next bar e

Displacement evolution

Force vector

$$\mathbf{f}_{n+1} = \mathbf{f}_{n+1}\left(\dot{\mathbf{u}}_{n+1}\right) \approx \mathbf{f}_{n+1}\left(\dot{\mathbf{u}}_{n}\right)$$

